HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2025)

Student No.: 12 Name: _____ Marks: / 9

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Let $g(x) = 3x^2 - 6kx + 24x + 3k^2 - 24k + 55$, where k is a positive constant. Denote the vertex of the graph of $y = g(x)$ by R.

   (a) Using the method of completing the square, express the coordinates of R in terms of k.

   (b) The graph of $y = h(x)$ is obtained by translating the graph of $y = g(x)$ rightwards by 6 units and then downwards by 10 units. Denote the vertex of the graph of $y = h(x)$ by S. Let T be a point such that the orthocentre of $\triangle RST$ is the origin.

      (i) Express the coordinates of T in terms of k.

      (ii) The coordinates of the point U are $(-5, 5)$. Let V be a point lying on RS such that TV is perpendicular to RS. It is given that S, T, U and V are concyclic. Find k.

(9 marks) (DSE-2025 PI Q18)

I can finish _____ questions

Answers written in the margins will not be marked.

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